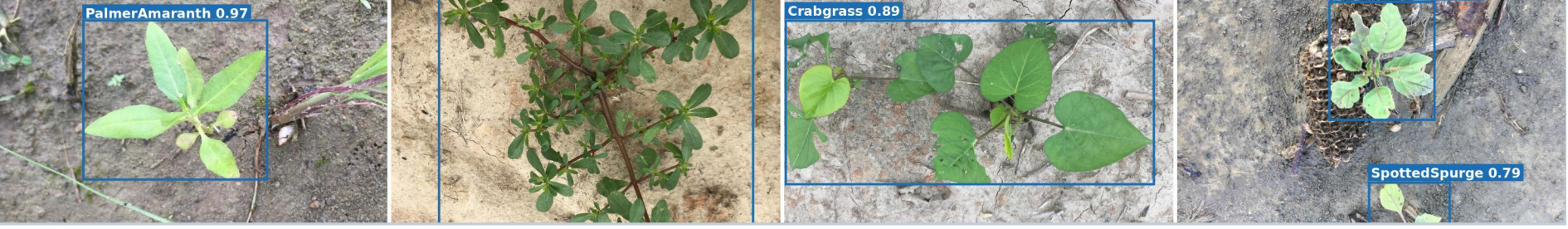
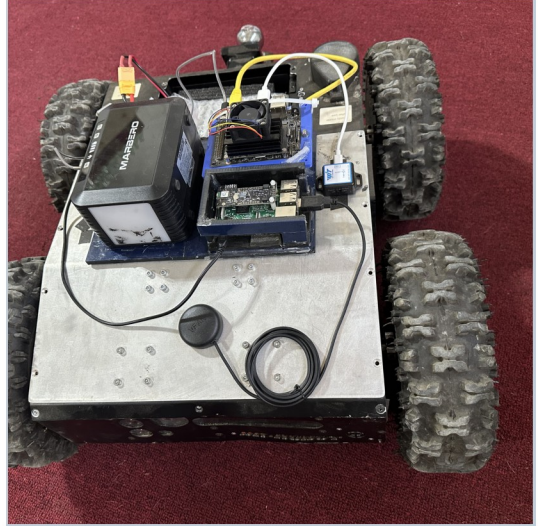


One platform where agents collect their own data and train on it, month after month, so the model compounds without anyone in the room. We built it, ran it unattended for fifteen rounds, and measured what each part of that loop is actually worth.

Weed Detection


Two agents work the corpus unasked: one harvests imagery and audits it, the other trains a detector small enough to ride the robots. More than 150,000 labelled images.

Rover


A four-wheel skid-steer the width of a crop row, carrying a Jetson, a Raspberry Pi and its own battery. Camera, GPS and two inertial units stream live over one API key.

Laser Weed Control Robot

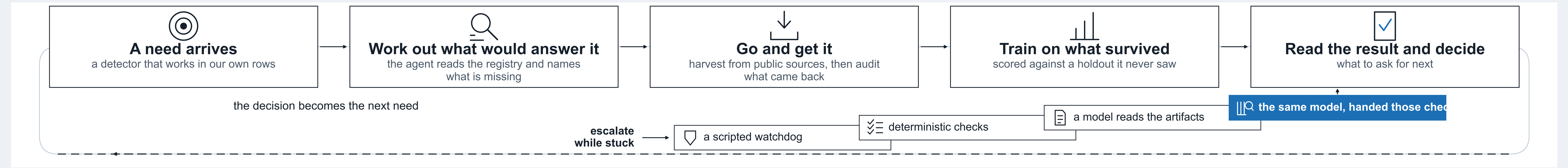

A twin-laser weeder. A YOLO detector finds the weed; two galvanometer lasers aim and fire, and it dies where it stands. No chemical, nothing touches the crop beside it.

Humanoid and Quadruped


A humanoid and a quadruped, out on the farm to record where wheels do not go. Their project takes the ingest contract the Rover uses; any session they open is governed.

What Each Project Is, and What It Is For

Figure 1. Every project on the platform opened the same way, same ingest contract, same agents train and evaluate it.



Agent Work Flow Diagram

Figure 2. A need arrives, the agent works out what data would answer it, fetches it, trains on what survives the audit, scores, decides what to ask next. Stuck, it climbs the ladder. Dashed: the decision is recorded, nothing reads it back.

ONE PLATFORM TAKES ANY ROBOT, ANY DATASET

OPEN WEIGHTS, AND A LADDER WHEN STUCK

AGENTS RAN THE LOOP, NOBODY IN THE ROOM

MORE DATA STOPPED HELPING (CONT.)

- agentAI is one platform for every lab robot
- A robot, drone or sensor rig is a project with its own keys, people and locked dataset registry. Agents harvest, filter, label and train against it. Adding a robot is a configuration change.
- The Rover and the laser robot both report in
- The Rover is a four-wheel skid-steer streaming camera, GPS and two inertial units. The Laser Weed Control Robot carries two galvanometer lasers on independent channels, a down camera over the work zone and a forward camera.
- A drive becomes a dataset as it opens
- The platform registers a vehicle's session as a governed dataset while the drive runs. Frames and sensor batches arrive over one API key. The laser robot sends no position.
- Any data type, any field, one project each
- Open a project, upload, and get an automatic diagnosis before choosing what to do next. Weed detection is one project here; so are the Rover and the laser robot.

Grounded recall separates the reviewers

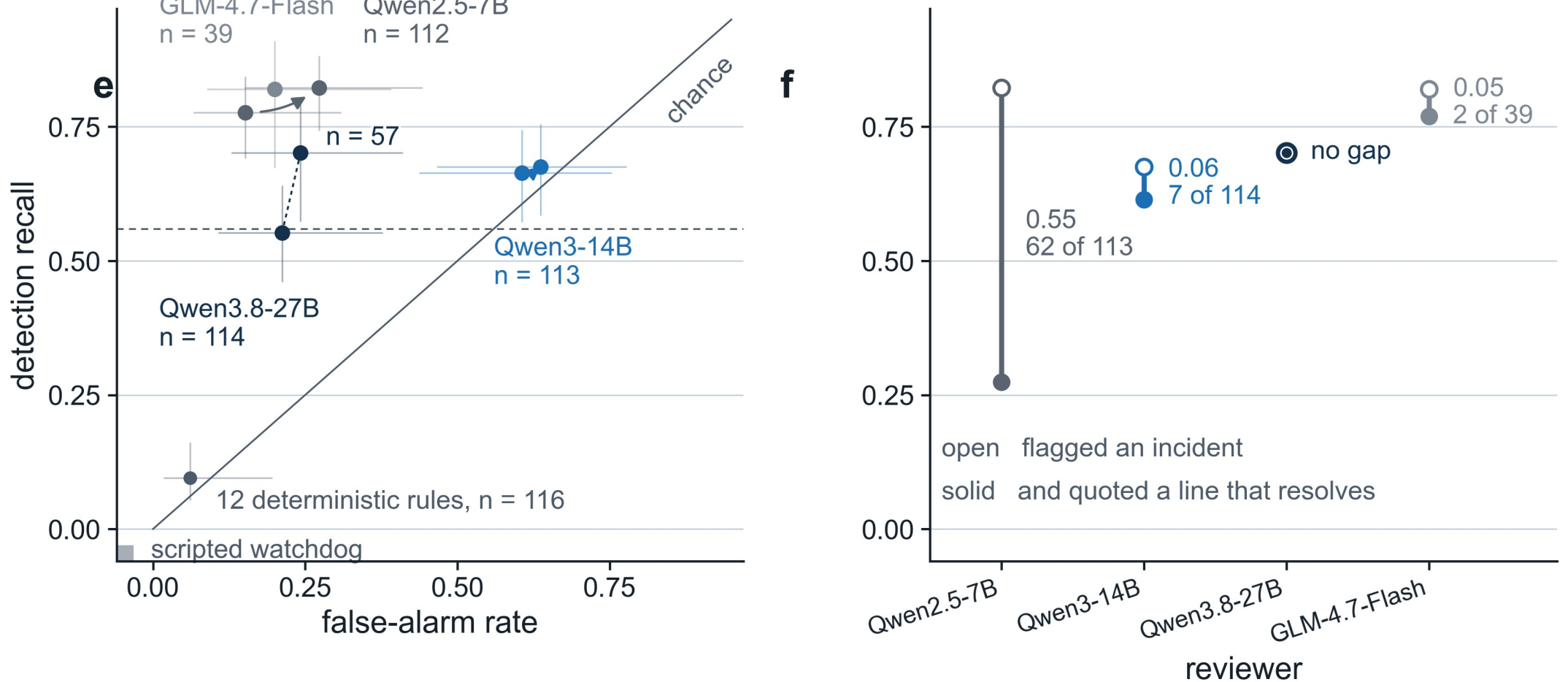


Figure 3. (e) Recall and (f) grounded recall on one frozen corpus of real incidents from our own record, scored by our own scorer. Tail: the model on the artifacts with the twelve checks' findings removed; head: the same model handed them. The dotted line, no arrowhead, marks a pair whose ends were scored on different cases -- the 27 B's head stands on 57 incidents against its tail's 114.

Escalating is worth it, and what it buys is grounding rather than size: handing the cheap tier's own findings to the model is worth +0.149 recall to the 27 B and +0.012 to the 14 B.

A cheap reviewer, and a bigger one above it when it is stuck

Review runs in tiers: a scripted watchdog, twelve deterministic checks, a model reading the artifacts with those checks hidden, then the same model handed them. On 149 frozen cases the scripted watchdog returned no decision at all.

The premise was compounding, and we ran it unattended

One agent collects, one trains, and next month's model should beat this month's. Fifteen rounds ran against the locked registry with nobody in the room, every round reported success, and the loop has been off since 2026-09-10.

MORE DATA STOPPED HELPING

More harvested data cost accuracy; we measured it

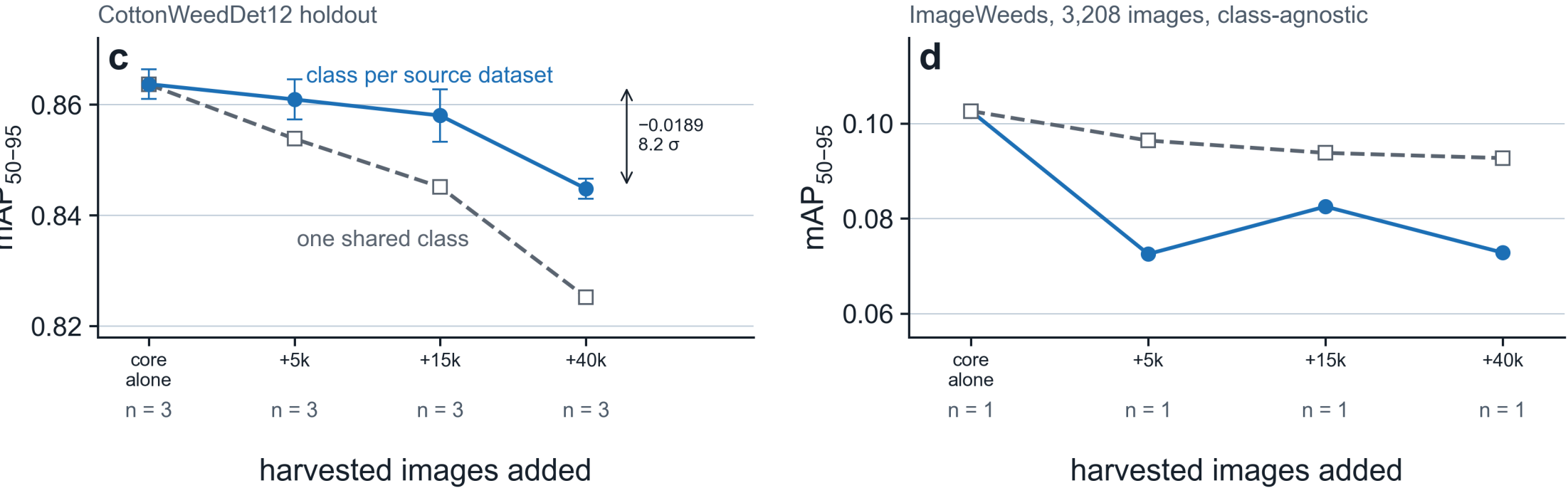


Figure 4. (c) Three seeds per rung, mean and 1 s.d., on the 1,977-image holdout -- which is also each run's validation set, so each mark is that run's best epoch. (d) Same eight checkpoints, unseen weed dataset, one seed per rung. Both exams fall.

Starting fresh instead of from the last round's weights recovers +0.0287, 5.1 times this recipe's seed spread. That is the change the next campaign makes.

- The recipe was not why more data stopped helping
- More data arrived; the detector did not improve. Augmentation, replaying known species, completing labels: each left the same gap. What stood was our auto-labeller's boxes.
- A vision model can judge a dataset; a chat model cannot
- Professor Zhang's rule: discard what sits too far from data we trust. Shown a candidate box, Gemma 4 rejects none of the false ones, so the test is DINOv2 geometry. Fifty-one datasets out, retrained, accuracy did not come back.
- The unit of curation has to match the unit of noise
- An auto-label error is one wrong box in an otherwise fine image, so a dataset-level filter cannot see it. The per-box tools are written, not re-measured. That is the open front.
- A brain's falsifiable hypothesis, and it was wrong
- Harvested boxes carry a class taken from the website they came from. A brain blamed that, wrote its falsifying condition first, and a human approved. Four runs falsified it. The real cost was density: nine boxes an image against under two.